

Due: December 16, 2025, 17:00 (Room R1102)

Important Notice: You must print this take-home quiz and **write your answers by hand with a pen.**

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Q1 Figure	Q2 Figure

Q1. (30 pts) Explain Breadth-First Search (BFS) on the graph and provide the BFS traversal order for the graph shown in Q1 Figure.

A1: BFS 是一種 graph 的遍訪演算法。BFS 從指定的 起始點 開始，先拜訪與起始點最近的所有相鄰節點，再一層一層的往外擴展。在遍訪過程中，會用一個 queue 來儲存接下來要拜訪的點，並用另外一個 陣列 記錄已經遍訪過的點，確保每個點只會被拜訪一次。

order (由左到右):

0 → 1 → 4 → 2 → 3 → 5 → 6 → 7

Q2. (30 pts) In tree traversal, one common method is inorder traversal. Please use inorder traversal to print the arithmetic expression represented by the expression tree in Q2 Figure, and then evaluate it to compute the final result.

A2: $3 \rightarrow x \rightarrow 4 \rightarrow + \rightarrow 10 \rightarrow \div \rightarrow 2 \rightarrow - \rightarrow 3$

故 $3 \times 4 + 10 \div 2 - 3$

$= 12 + 5 - 3$

$= 14$

Q3. (40 pts) A binary tree is a fascinating data structure with many variations, including binary search trees, AVL trees, red-black trees, complete binary trees, and max/min heaps. These variations can be classified as shape-based (structural constraints) or criteria-based (rules such as ordering). Choose one shape-based tree and one criteria-based tree, and provide a brief description of each.

A3:

Shape-based:

Binary Tree, 因為 Binary Tree 被定義為每個 node 最多只會有兩個 child, 且 node 之間沒有數值排序的限制。因此 Binary Tree 是屬 structural constraints 的樹。
任何

Criteria-based:

Binary Search Tree, 因為 Binary Search Tree 的每個 node 皆需滿足排序規則: 每個 left subtree 中所有 node 的值皆必需小於該 node, 而 right subtree 中所有 node 的值皆必需大於該 node, 此條件會套用到整棵樹, 因此是 criteria-based tree。